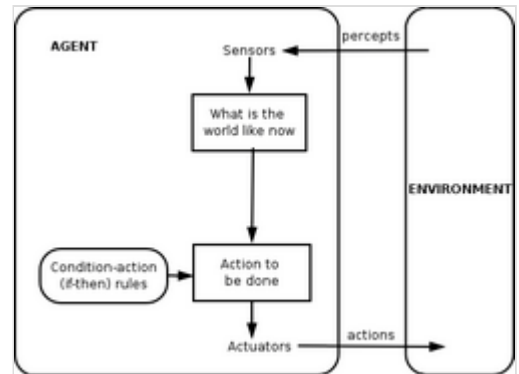


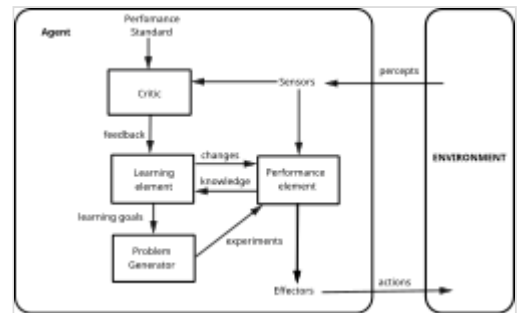
Multi-agent system

A **multi-agent system (MAS)** or "self-organized system" is a computational system composed of multiple interacting intelligent agents.^{[1][2][3]} Multi-agent systems can solve problems that are difficult or impossible for an individual agent or a monolithic system to solve.^[4] Intelligence may include methodic, functional, procedural approaches, algorithmic search or reinforcement learning.^[5] With advancements in large language models (LLMs), LLM-based multi-agent systems have emerged as a new area of research, enabling more sophisticated interactions and coordination among agents.^[6]



Simple reflex agent

Despite considerable overlap, a multi-agent system is not always the same as an agent-based model (ABM). The goal of an ABM is to search for explanatory insight into the collective behavior of agents (which do not necessarily need to be "intelligent") obeying simple rules, typically in natural systems, rather than in solving specific practical or engineering problems. The terminology of ABM tends to be used more often in the science, and MAS in engineering and technology.^[7] Applications where multi-agent systems research may deliver an appropriate approach include online trading,^[8] disaster response,^{[9][10]} target surveillance^[11] and social structure modelling.^[12]



Learning agent

Concept

Multi-agent systems consist of agents and their environment. Typically, research on multi-agent systems refers to software agents. However, the agents in a multi-agent system could equally well be robots, humans, or human teams, and may consist of combined human-agent teams.

Agents can be divided into types spanning simple to complex. Categories include:

- Passive agents^[13] or "agent without goals" (such as obstacle, apple or key in any simple simulation)
- Active agents^[13] with simple goals (like birds in flocking, or wolf–sheep in prey-predator model)
- Cognitive agents, with beliefs, desires, intentions, and commitments processed by logical, probabilistic, and neural network-based reasoning

Agent environments can be divided into:

- Virtual

- Discrete
- Continuous

Agent environments can also be organized according to properties such as accessibility (whether it is possible to gather complete information about the environment), determinism (whether an action causes a definite effect), dynamics (how many entities influence the environment in the moment), discreteness (whether the number of possible actions in the environment is finite), episodocity (whether agent actions in certain time periods influence other periods),^[14] and dimensionality (whether spatial characteristics are important factors of the environment and the agent considers space in its decision making).^[15] Agent actions are typically mediated via an appropriate middleware. This middleware offers a first-class design abstraction for multi-agent systems, providing means to govern resource access and agent coordination.^[16]

Characteristics

The agents in a multi-agent system have several important characteristics:^[17]

- **Autonomy:** agents are at least partially independent, self-aware, autonomous
- **Local views:** no agent has a full global view, or the system is too complex for an agent to exploit such knowledge
- **Decentralization:** no agent is designated as controlling (or the system is effectively reduced to a monolithic system)^[18]

Self-organisation and self-direction

Multi-agent systems can manifest self-organisation as well as self-direction and other control paradigms and related complex behaviors even when the individual strategies of all their agents are simple. When agents can share knowledge using any agreed language, within the constraints of the system's communication protocol, the approach may lead to a common improvement. Example languages are Knowledge Query Manipulation Language (KQML) or Agent Communication Language (ACL).

Decision-Making

Decision protocols in multi-agent systems refer to the structured rules and procedures that agents follow to reach collective decisions or agreements. Such protocols specify how agents share information, negotiate, and resolve conflicts, ensuring coordinated behavior and effective joint actions. Decision protocols can range from voting mechanisms to consensus-building algorithms, and they significantly influence the efficiency and reliability of multi-agent interactions.^[19]

System paradigms

Many MAS are implemented in computer simulations, stepping the system through discrete "time steps". The MAS components communicate typically using a weighted request matrix, e.g.

```
Speed-VERY_IMPORTANT: min=45 mph,
Path length-MEDIUM_IMPORTANCE: max=60 expectedMax=40,
Max-Weight-UNIMPORTANT
Contract Priority-REGULAR
```

and a weighted response matrix, e.g.

```
Speed-min:50 but only if weather sunny,  
Path length:25 for sunny / 46 for rainy  
Contract Priority-REGULAR  
note - ambulance will override this priority and you'll have to wait
```

A challenge-response-contract scheme is common in MAS systems, where

- First a "Who can?" question is distributed.
- Only the relevant components respond: "I can, at this price".
- Finally, a contract is set up, usually in several short communication steps between sides,

also considering other components, evolving "contracts" and the restriction sets of the component algorithms.

Another paradigm commonly used with MAS is the "pheromone", where components leave information for other nearby components. These pheromones may evaporate/concentrate with time, that is their values may decrease (or increase).

Properties

MAS tend to find the best solution for their problems without intervention. There is high similarity here to physical phenomena, such as energy minimizing, where physical objects tend to reach the lowest energy possible within the physically constrained world. For example: many of the cars entering a metropolis in the morning will be available for leaving that same metropolis in the evening.

The systems also tend to prevent propagation of faults, self-recover and be fault tolerant, mainly due to the redundancy of components.

Research

The study of multi-agent systems is "concerned with the development and analysis of sophisticated AI problem-solving and control architectures for both single-agent and multiple-agent systems."^[20] Research topics include:

- agent-oriented software engineering
- beliefs, desires, and intentions (BDI)
- cooperation and coordination
- distributed constraint optimization (DCOPs)
- organization
- communication
- negotiation
- distributed problem solving
- multi-agent learning^[21]
- agent mining
- scientific communities (e.g., on biological flocking, language evolution, and economics)^{[22][23]}

- dependability and fault-tolerance
- robotics,^[24] multi-robot systems (MRS), robotic clusters
- multi-agent systems also present possible applications in microrobotics,^[25] where the physical interaction between the agents are exploited to perform complex tasks such as manipulation and assembly of passive components.
- language model-based multi-agent systems^[6]

A MAS involves more than just the design of an intelligent system. It also provides insights and understanding about interactions among humans, as they organize themselves into various groups, committees, societies, and economies in order to improve their lives. For example, economists have been studying multiple agents for more than two hundred years, ever since Adam Smith in the eighteenth century, with the goal of being able to understand and predict economies. Economics provides ways to characterize masses of agents. and these are useful for DAI. But in return, DAI provides a means to construct artificial economies that can test economists' theories before, rather than after, they are applied.

Frameworks

Frameworks have emerged that implement common standards (such as the FIPA and OMG MASIF standards).^[26] These frameworks e.g. JADE, save time and aid in the standardization of MAS development.^[27]

Currently though, no standard is actively maintained from FIPA or OMG. Efforts for further development of software agents in industrial context are carried out in IEEE IES technical committee on Industrial Agents.^[28]

With advancements in large language models (LLMs) such as ChatGPT, LLM-based multi-agent frameworks, such as CAMEL,^{[29][6]} have emerged as a new paradigm for developing multi-agent applications. Recent work has shown that such debate-oriented systems vary in their orchestration (e.g., discussion paradigms^[30]). The MALLM framework is used to systematically evaluate possible configurations of frameworks.^[31]

Applications

MAS have not only been applied in academic research, but also in industry.^[32] MAS are applied in the real world to graphical applications such as computer games. Agent systems have been used in films.^[33] It is widely advocated for use in networking and mobile technologies, to achieve automatic and dynamic load balancing, high scalability and self-healing networks. They are being used for coordinated defence systems.

Other applications^[34] include transportation,^[35] logistics,^[36] graphics, manufacturing, power system,^[37] smartgrids,^[38] and the GIS.

Also, Multi-agent Systems Artificial Intelligence (MAAI) are used for simulating societies, the purpose thereof being helpful in the fields of climate, energy, epidemiology, conflict management, child abuse,^[39]

Some organisations working on using multi-agent system models include Center for Modelling Social Systems,^[40] Centre for Research in Social Simulation,^[41] Centre for Policy Modelling, Society for Modelling and Simulation International.^[39]

Vehicular traffic with controlled autonomous vehicles can be modelling as a multi-agent system involving crowd dynamics.^[42]

Hallerbach et al. discussed the application of agent-based approaches for the development and validation of automated driving systems via a digital twin of the vehicle-under-test and microscopic traffic simulation based on independent agents.^[43] Waymo has created a multi-agent simulation environment Carcraft to test algorithms for self-driving cars.^{[44][45]} It simulates traffic interactions between human drivers, pedestrians and automated vehicles. People's behavior is imitated by artificial agents based on data of real human behavior.

See also

- Comparison of agent-based modeling software
- Agent-based computational economics (ACE)
- Artificial brain
- Artificial intelligence
- Artificial life
- AI mayor
- Black box
- Blackboard system
- Complex systems
- Discrete event simulation
- Distributed artificial intelligence
- Emergence
- Evolutionary computation
- Friendly artificial intelligence
- Game theory
- Hallucination (artificial intelligence)
- Human-based genetic algorithm
- Hybrid intelligent system
- Knowledge Query and Manipulation Language (KQML)
- Microbial intelligence
- Multi-agent planning
- Multi-agent reinforcement learning
- Pattern-oriented modeling
- PlatBox Project
- Reinforcement learning
- Scientific community metaphor
- Self-reconfiguring modular robot
- Simulated reality
- Social simulation

- Software agent
- Software bot
- Swarm intelligence
- Swarm robotics

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